

ECHO MUSIC PLAYER

Software Requirement Specifications (SRS)

Version 1.0

PROJECT WORK PHASE - I (ECS799)

Degree

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Approved by:**Project Guide Name: Mr. Rupal Gupta****Signature:**

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1 Introduction

The Echo Music Player is an Android Music Player Application that will be one of its kind free music player out there with all new looks, functionalities and features. It will enable users to play their music or audio files in all a new style and with added features, low space consumption and reliability.

1.1 Purpose

This SRS document provides a high level description and detailed requirements necessary for the development of the intended Echo Music Player Android App and will thus form a guideline for designing and developing the same end product.

1.2 Scope of the Work

Echo Music Player is intended to become one of a kind free general purpose Music Player Android Application.

Definitions, Acronyms, and Abbreviations

Abbreviation	Description
DFD	Data Flow Diagram
IDE	Integrated Development Environment
OS	Operating system
RAM	Random Access Memory
SRS	Software Requirement Specifications

1.3 References

a) Books

- Android Development with Kotlin by Igor Wojda
- Software Engineering Principles and Practices by Roger Pressman

b) Websites

- www.edx.com
- www.stackoverflow.com

S#	Reference Details	Owner	Version	Date
1.	Project Synopsis & Documentation	Manish Gupta	1.0	25-09-2020

2 Overall Description

The Echo Music Player is an Android Music Player Application that will be one of its kind free music player out there with all new looks, functionalities and features. It will enable users to play their music or audio files in all a new style and with added features, low space consumption and reliability. The only pre-requisite for its effective usage is that app is downloaded on the device and the audio or music files that the user intends to play here must be present on his/her device or external storage physically beforehand for functioning properly, the list of which will be dynamically updated each time user launches the app in case the user adds new media files to the device or its storage whether internal or external, however it is recommended to the users of these app to use only licensed or authorized music, audio or other similar creative works and play the same to avoid legal implications for the same, the project can nowhere be held responsible since it doesn't encourage software, music, artistic or creative work piracy in any form. Thus, make sure you're aware of this and authorized to play what you play here in this project.

2.1 Product Perspective

Echo Music Player is developed to be amongst the best of its kind general purpose Music/Audio player available in the marketplace as:-

- Low on Space/Size Requirements.
- No Premium memberships/Registrations and in compliance with copyrights thus no violations.
- No hidden charges thereby in compliance with free and fair use policies.
- No discriminative behavior amongst users as designed for a non-profitable purpose.
- Great UI when compared to its size and serves its purpose with a great UX in this niche.
- No data collection or hidden monitoring thereby completely safe and secure for users.



2.2 Product Functions

The main purpose for the development of Echo Music Player is to design a free to use general purpose Music/Audio player for Android devices with a minimal end on device space consumption thereby targeting and being backward compatible for Android Devices working with a low end on computational capabilities, thereby serving the purpose and being in ultra-modern look and UI of a kind, become a best choice for such users and not only that, also winning the hearts of users with modern end devices serving the purpose without legally infringing or violating anything.

- Shake to Change Functionality : Give your phone a shake to change the playing track
- Supports All types of Audio Files

- Houses a Powerful attention-capturing Open Source Equalizer
- In sync with Headset/Bluetooth Controls
- The App must be User Friendly – Great UI and UX
- Create Custom Playlists by Adding your Tracks to Favorites
- Smart Remembering feature automatically applies previously Selected Controls

2.3 User Characteristics

Since the end product will be a free to use general purpose Music Player Android Application, all the end users will be characterized by the system with same set of authorization and permissions as we won't be collecting any sort of user data from the Application, all the users will be having relatively the same i.e. equivalent rights in the App, thereby promoting fairness for all users involved. All users will be thereby treated as equals, whether the developer, tester, designer, manager or any random user of the App, all have same level of Authority within the App. And, A General user in this App would be able to Play Songs/Audio files fetched from his device internal/external storages, mark/unmark them as favorites, activate Shake to Change the song functionality within the app and much more.

2.4 General Constraints

- a) RAM : at least 256 MB of RAM
- b) Hard Disk : at least 10 MB of Disk Space on Hard Disk
- c) OS : Android Operating System

2.5 Assumptions and Dependencies

Basically, “Echo – Music Player” application needs a Mobile device running on Android OS v4.0.4 or above and after Installation, it needs access to some permissions like Storage and Audio settings to function desirably, these permissions are optional though recommended as the App may misbehave without granting them or may crash abruptly. Hence, it's an assumption to use an Android Device and permissions are Dependencies that are subject to change as per Google Android Play Store Policies.

3 Specific Requirements

In this project, we aim to build a general purpose Music Player Android application with the following functionalities:

- A glossy Splash screen and a custom Navigation drawer with app logo section at the top along with links to all other fragments of the App.

- An 'All songs' screen, the home screen of the app, mentioning songs fetched from device and external storage media on device.
- The app should be able to fetch and play possibly all sorts of audio files.
- A 'Favorites' screen displaying lists of song marked as favorites earlier.
- A 'Settings' screen displaying app configuration settings.
- An 'About us' screen displaying more information about the project developer.
- A 'Now playing' screen with powerful equalizer and other necessary controls and options.
- A 'Now playing' mini bar at the bottom with name of the track playing and play or pause feature. This would appear if the user has transitioned from 'Now playing' screen to either the 'All songs' screen or the 'Favorites' screen without pausing the track.
- Background play. The app will continue playing the track if the app gets minimized without the music being paused.

3.1 Play Music

3.1.1 Description

This function will first fetch a list of all audio files present in the device then will display that list to the user and then plays the selected music file that the user intends to listen to.

3.1.2 Inputs

Fetches a list of songs in the device using a user-defined function and user selected audio data file becomes an input to media player.

3.1.3 Processing

The function calls intrinsic feature that displays list of all music files on the device and its external storage on the screen and once the user selects one of them, this function transfers the selected song data to the media player and the custom designed Song Playing Screen where it gets played.

3.1.4 Outputs

The selected song gets played on the custom designed screen and in the background too even if the app is minimized unless the running app service is killed from the background and System UI.

3.1.5 Error Handling

None

3.1.6 Test Cases for Play Music

Test Case ID	Test Title	Test Case Description	Pre-condition, if any	Expected Results
TC001	Run & Play	To check Songs fetched from device and play selected song in that list	None	All Audio files listed & selected item played
TC002	Seek/Pause	Seeking/Pausing the selected song	Song must be played first	App must seek the song or change song state
TC003	Stop	Stopping the Player and App	App is running and Song is playing	The app services stops till next run

3.2 Favorites

3.2.1 Introduction

This function will add a selected playing song to user favorites which can be later played from favorites screen and also can be used to remove from favorites which will be added or removed in a dynamic favorite database in the app data (Echo Database).

3.2.2 Inputs

Currently Playing Song Data.

3.2.3 Processing

This function intrinsically calls a feature that on being invoked creates a dynamic database i.e. Favorite Database (Echo Database). for Favorite marked songs in the App data and selected values reappear in even next running cycles of the app unless disabled by the user.

3.2.4 Outputs

The Favorite marked song is stored in dynamic Favorite database (Echo Database). within the App data and can be seen thereby on the favorites screen once done, any changes are reflected there and are observed intact in subsequent App running cycles.

3.2.5 Error Handling

None

3.2.6 Test Cases for Favorites

Test Case ID	Test Title	Test Case Description	Pre-condition, if any	Expected Results
TC004	Added to Favorites	Song marked as favorites	None	Song Added to Favorites
TC005	Removed from Favorites	Song unmarked as favorites	Song must be favorite first	Song Removed from Favorites

3.3 Saved Controls

3.3.1 Introduction

This function will add a user selected control to App data that is used to shuffle, loop or change song by shaking device features which will be stored in App's Shared preferences.

3.3.2 Inputs

State of Control features like shuffle, repeat, accelerometer usage and looping preferences.

3.3.3 Processing

This function intrinsically calls a feature that on being invoked adds user selected controls to app's shared preferences which stores them in the App data and selected values reappear in even next running cycles of the app unless disabled by the user.

3.3.4 Outputs

The selected control features gets stored in App data in the shared preferences and are observed intact in subsequent running cycles of the App.

3.3.5 Error Handling

None

3.3.6 Test Cases for Saved Controls

Test Case ID	Test Title	Test Case Description	Pre-condition, if any	Expected Results
TC006	Loop	Looping turned on/off	None	Looping gets enabled/disabled
TC007	Shuffle	Shuffling turned on/off	None	Shuffling gets enabled/disabled

TC008	Shake & Change	Shaking device to swap Song turned on/off	None	Accelerometer Sensitivity gets enabled/disabled for swapping songs
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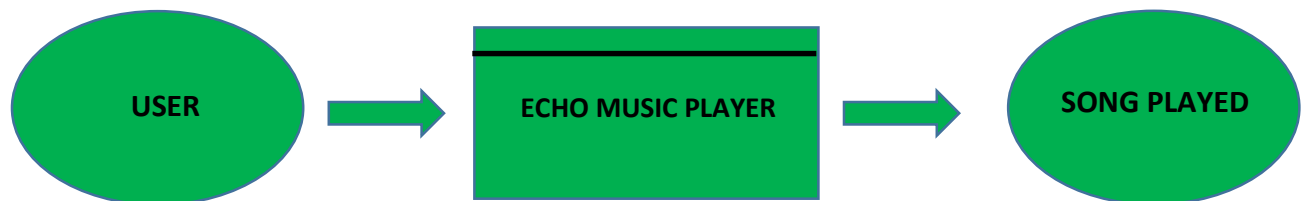
3.4 Design Constraints

- Permissions: App permissions are Constraints that are subject to change as per Google Android Play Store Policies.
- Space: Hardware Limitation, we need to minimize the App Space the best we can in order to make it the best Lightweight Music Player Application out there.
- Programming Language: Kotlin, with itself and SQLite for Backend and Shared Preferences.
- Copyright: Nothing within or outside the app, internally or externally used must violate or become a cause or case of Software, Literary or Music Copyright Infringement, hence everything used inside the app and externally must promote free and fair use policy and shouldn't violate any copyrights.

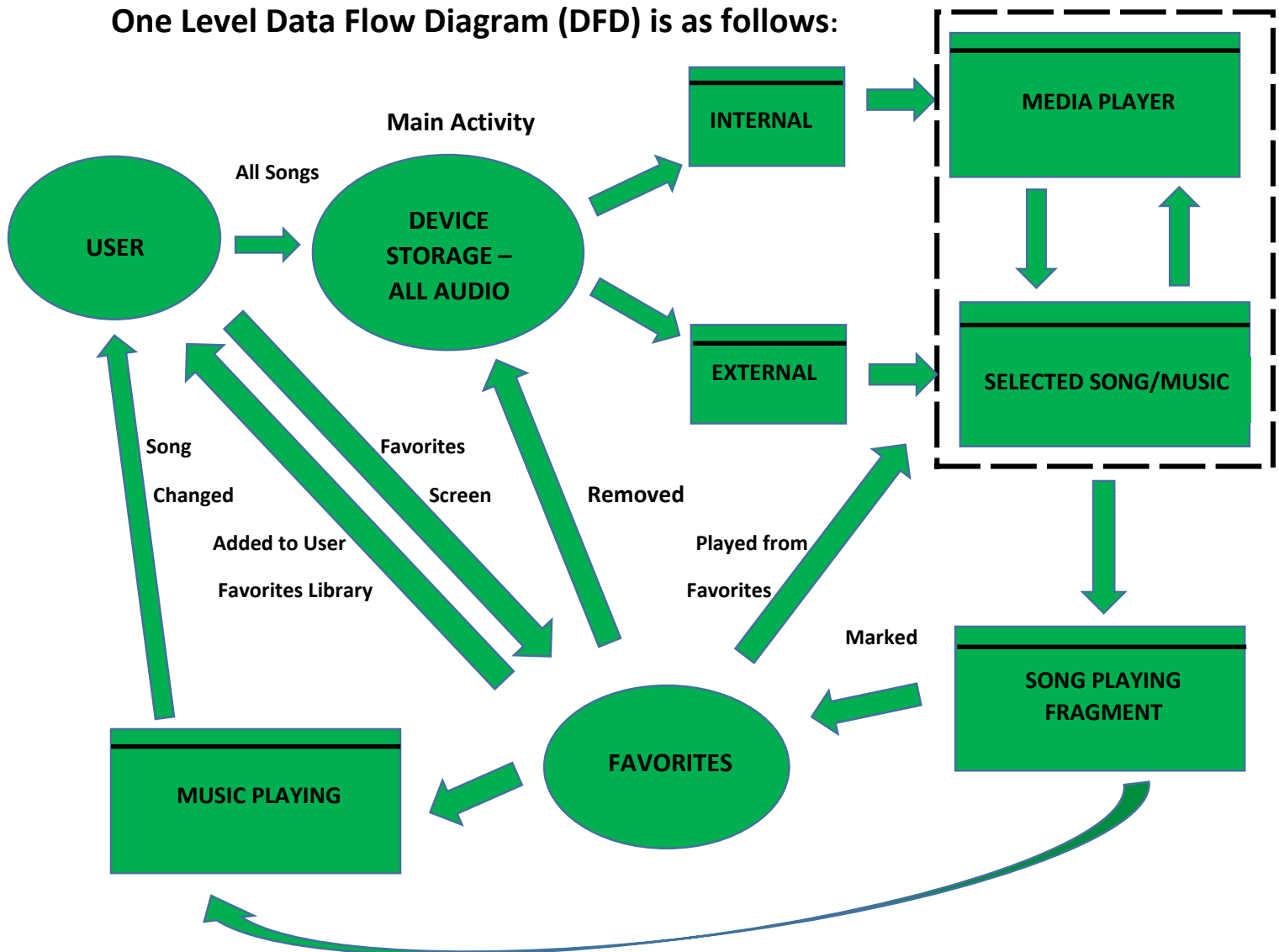
Appendix A

Data Flow Diagram (DFD)

Zero Level Data Flow Diagram (DFD) is as follows:



One Level Data Flow Diagram (DFD) is as follows:



Appendix B

Data Dictionary (DD)

FavoriteTable (in FavoriteDatabase) (EchoDatabase)

Fields	Data type	Description
SongID	Integer	Song ID
SongTitle	Text	Song Name
SongArtist	Text	Artist Name
SongPath	Text	Song's Physical Location

End of Project SRS Document



**Thank
You!!!**